

1. which of the following is correct about Plasma Membrane

☐ providing a barrier for the cell

☒ all are correct

☐ It is lipid/protein/carbohydrate complex,

☐ containing transport and signaling systems

2. Which of the following are correct about amino acids

☐ Non is correct

☒ Contains amino group and carboxyl group function groups

☐ 30 commonly amino acids are present in protein biosynthesis

☐ Individual amino acids in protein connected by glycoside bond.

3. Biochemical studies

- ☐ have illuminated many aspects of health
- ☐ have illuminated many aspects of disease,
- ☒ a and b
- ☐ Non

4. All the following have 18 carbon atoms except ?

☐ Linoleic acid

☐ Stearic acid

☐ Linolenic acid

☒ Arachidonic acid

5. which of the following statement about metabolism is correct:

☒ all are correct



Total sum of the chemical reaction happening in a living organism



Anabolism- energy requiring biosynthetic pathways



Catabolism- degradation of fuel molecules and the production of energy for cellular function

6. In amylopectin the intervals of glucose units of each branch is ?

☐ 40–50

☒ 24–30

☐ 30–40

☐ 10–20

7. Maltose can be formed by hydrolysis of ?

☐ Glycogen

☐ All of these

☒ Starch

☐ Dextrin

8. The principle of biochemistry are as follow:



Certain important reactions E.g. Glycolysis is found in almost all organisms.



all are correct



Living processes contains thousands of chemical. Precise regulation and integration of these reactions are required to maintain life



All organisms use the same type of molecules: CHO, proteins, lipids & nucleic acids.

9. The glucose is a ?

☒ hexose

☐ heptose

☐ pentose

☐ sucrose

10. Which of the following is correct about the Cell

- ☐ Many can not be seen with the naked eye
- ☐ Smallest living unit of an organism
- ☐ Grow, reproduce, use energy, adapt, respond to their environment
- ☒ All are correct

11. The polysaccharide found in the exoskeleton of invertebrates is ?

☐ glycogen

☒ Chitin

☐ Chondroitin sulphate

☐ Cellulose

12. Cerebrosides contain all the following except ?

☐ Galactose

☒ Sulphate

☐ Fatty acid

☐ Sphingosine

13. Which of the following is the smallest carbohydrate ?

☐ Ribose

☐ Glucose

☐ None of them

☒ Glyceraldehyde

14. Sphingomyelin contains ?

☐ plasmalogen

☐ Glycerol

☒ Sphingosine

☐ Serine

15. Which is the following is aldotriose ?

☒ Glyceraldehyde

☐ Cellulose

☐ ribose

☐ Glucose

16. which of the following is correct about Mitochondria



Surrounded by a double membrane with a series of folds called cristae.



Functions in energy production through metabolism.



All are corrects



Contains its own DNA, and is believed to have originated as a captured bacterium

17. Carbohydrate in cerebrosides is
?

☐ Maltose

☐ Fructose

☐ Glucose

☒ Lactose

18. The number of double bonds in arachidonic acid is ?

☐ 1

☒ 4

☐ 2

☐ 3

19. Which carbohydrate cannot be digested and absorbed ?

☒ Cellulose

☐ b-Glycogen

☐ c-Starch

☐ d-All of these

20. Amylopectin has ?

☐ β - 1-4 and β - 1-6 linkage

☐ α - 1-2 linkage

☒ α - 1-4 and α - 1-6 linkage

☐ β - 1-2 linkage

21. Two sugars which differ from one another only in configuration around a single carbon atom are termed ?

☐ Optical isomers

☐ Stereoisomers

☐ Anomers

☒ Epimers

22. Which one of the following is phospholipid ?

☒ All of them

☐ Cephalin

☐ Lecithin

☐ Sphingomyelin

23. What is the definition of biochemistry:

☐ all are correct



the ways these molecules interact to form cells, tissues and whole organism



Studying the structure and behavior of the complex molecules found in biological material



Biochemistry is the application of chemistry to the study of biological processes at the cellular and molecular level

24. The nitrogenous base in lecithin is ?

- ☐ Serine
- ☐ Ethanolamine
- ☐ Inositol
- ☒ Choline

25. function of protein are as follow:

☒ All are correct

☐ antibodies

☐ enzymes

☐ transport proteins

26. Nutritional polysaccharides are
?

☐ Starch and cellulose

☐ Starch and chitin

☐ Starch and glucose

☒ Starch and glycogen

27. Which of the following is a polyunsaturated fatty acid ?

☒ Linoleic acid

☐ Oleic acid

☐ Palmitoleic acid

☐ Palmitic acid

28. What are the objective of Biochemistry

- ☐ Know what biochemistry is and its principle.
- ☐ Know the components of a cell and its major types of bio-molecules.
- ☐ Understand how the role of cell organisation and different types of chemical reactions involved in maintaining high degree of internal order.
- ☐ All are correct

29. Glycosphingolipids are a combination of ?

☐ Glycerol with galactose

☒ Ceramide with one or more sugar residues

☐ Sphingosine with galactose

☐ Sphingosine with phosphoric acid

30. Which of the following is not an unsaturated fatty acid?

☒ Palmitic acid

☐ Palmitoleic acid

☐ Linoleic acid

☐ Oleic acid